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Supplemental information

A highly efficient heat-driven thermoacoustic cooling system

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Supplemental Experimental Procedures

Details of Experiment Configuration

The experiment setup of the present system is illustrated in Figure S1, and the main geometric dimensions are listed in Table S1 in detail. The diameter of the engine is smaller than that of cooler for the better energy-flow field synergy and reducing loss. The HHX and CHX are the plate-fin type, whereas the AHXs of engine and cooler are the shell-tube type to realize better heat transfer between the working gas and cooling water. In particular, the AHX of cooler employs a double-pipe configuration, in which a large tube (3.2 mm in diameter) and four small tubes (1 mm in diameter) are welded together. Accordingly, the heat transfer area is enlarged by several times to enhance heat transfer. The geometric parameters are optimized by the numerical method which will be described subsequently.

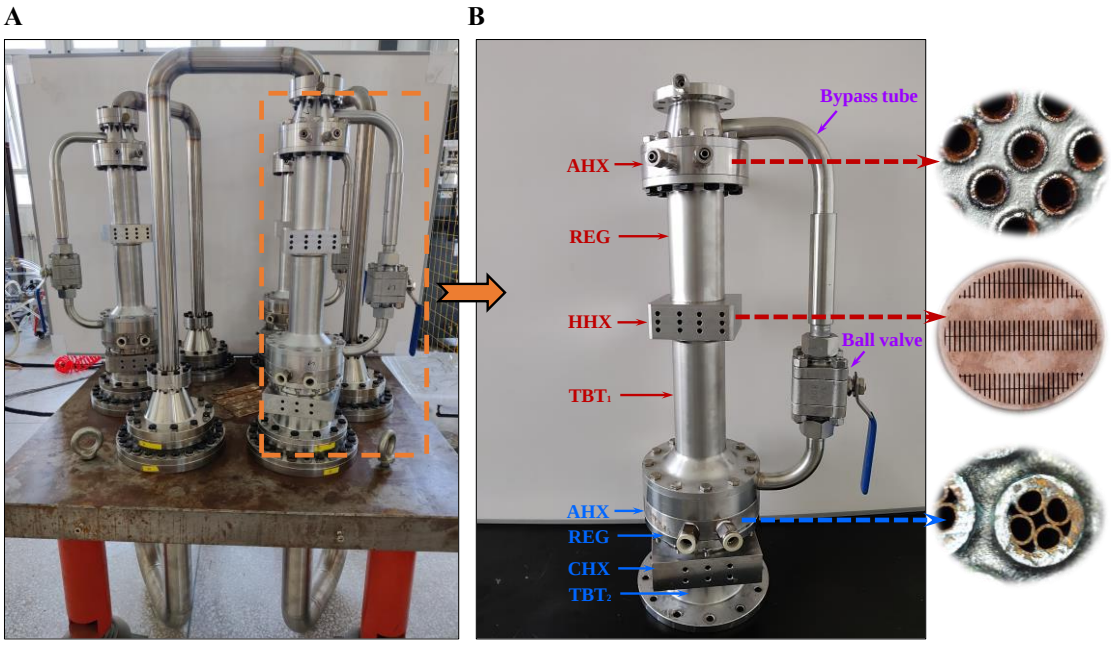


Figure S1. Experiment setup of the present heat-driven thermoacoustic refrigerator with bypass configuration. (A) The overall system including three subunits. **(B)** A subunit of the system.

Table S1. Main geometric dimensions of the present experimental setup.

Unit	Component	Diameter (mm)	Length (mm)	Other dimensions
Engine	AHX	65.5	35	Shell-tube type, 1.1 mm in hydraulic diameter, 425 in tube number
	REG	65.5	150	Mesh screen, 80% in porosity, 50 μm in wire diameter
	HHX	65.5	30	Plate-fin type, 7.2 mm in fin length, 1.5 mm in fin width, 18.6% in porosity
Cooler	AHX	110	35	Shell-tube type, 0.5 mm in hydraulic diameter, 15% in porosity
	REG	110	35	Mesh screen, 80% in porosity, 50 μm in wire diameter
	CHX	110	35	Plate-fin type, 9 mm in fin length, 1 mm in fin width, 22.5% in porosity
Others	TBT_1	65.5	178	
		65.5 – 110	30	
	TBT_2	110	55	
	Bypass tube	21.4	652	
	Ball valve	20	110	
	Expansion chamber	150	32	376.8 mm^3 in volume
	Liquid resonator	68	--	Water, 5 kg in mass
	Connection tube	40	973	

In the experiment, there are five parameters involved for measurements: the operating frequency, mean pressure, oscillating pressure, temperature, and power. The temperature contains heating, ambient and cooling temperatures, which are measured by installing thermocouples into the HHX (K-type), cooling water pipe (T-type) and CHX (T-type), respectively. The power is given by electricity for heating: after reaching equilibrium, for the HHX, the heating power given is measured the electric power; regarding the CHX, according to the energy-conservation law, the cooling power is equal to the imposed heating power, i.e., the given electric power. The main information of the measurement sensors is given in Table S2, and the photograph of

these sensors are shown in Figure S2.

Table S2. Main information of the measurement sensors.

Measurement parameter	Uncertainty	Details
Operating frequency	±1%	Dynamic pressure sensor from PCB Piezotronics
Oscillating pressure	±1%	Dynamic pressure sensor from PCB Piezotronics
mean pressure	±0.25%	Static pressure sensor (JYB-KO-HVG)
Temperature	±0.5 K	T-type thermocouple
	±0.5 K	K-type thermocouple
Electric power	0.1% × display value + 0.1% × measurement range	Electric power meter

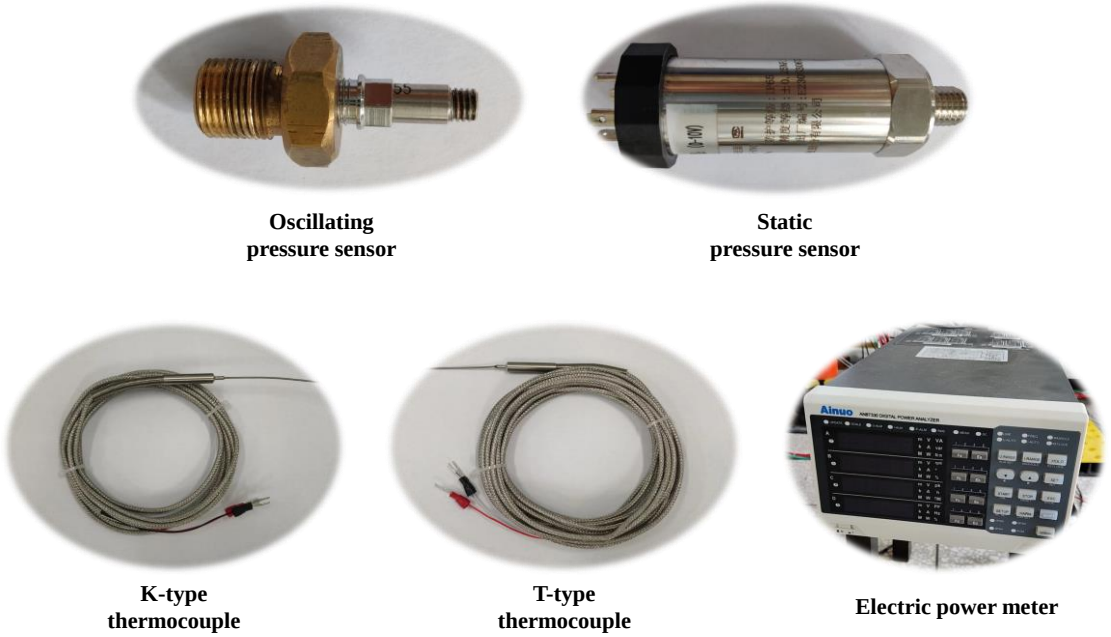


Figure S2. Photographs of measurement sensors.

Post Processing of Data

The COP is obtained by the ratio of cooing power and heating power, as follows

$$\mathrm{COP} = \frac{\dot{Q}_c}{\dot{Q}_h} \tag{S1}$$

In this study, both of $\dot{Q}_c$ and $\dot{Q}_h$ are tested by direct electric heating, thus they can be directly measured by the electric power meter. The heating power $\dot{Q}_h$ represents the

thermal energy absorbed by the HHX, in which the heat leak of insulation materials is not considered. This is more reasonable to understand the intrinsic performance of the proposed system, because the heat leak of insulation materials can be changeable depending on the materials used for insultation (i.e., it could reach a low value near zero with the improvement of insulation materials). The heat leak $\dot{Q}_{leak,m}$ of insulation materials can be estimated by

$$\dot{Q}_{leak,m} = \dot{Q}_{leak,total} - \dot{Q}_{loss,in} \tag{S2}$$

where $\dot{Q}_{leak,total}$ and $\dot{Q}_{loss,in}$ denote the total heat leak and intrinsic internal heat loss. The total heat leak is obtained by the experiment: At the static situation (i.e., the system does not operate thus without oscillation), when the system reaches thermal equilibrium at the given heating temperature, the total heat leak is equal to the heat added (i.e., the heating power displayed), thus can be measured by the electric power meter.

For engine, the intrinsic internal heat loss mainly occurs in REG and TBT_1, because they are the only two components with enormous axial temperature gradient in engine. Our tests indicate that the natural heat convection in the REG (due to the top-placed hot end and the bottom-placed cold end) has negligible impact in the intrinsic heat loss of the system at the non-oscillating condition, thus is not considered in this study. Therefore, in the REG, the intrinsic internal heat loss contains three parts of heat conduction: the heat conduction of solid wall, mesh screen, and working gas. According to Fourier's law, it can be expressed by

$$\dot{Q}_{loss,in}\big|_{REG} = \left[k_{sw}A_{sw}\frac{dT}{dx} + \beta k_{ms}A_{ms}\frac{dT}{dx} + k_g A_g \frac{dT}{dx}\right]\Big|_{REG} \tag{S3}$$

where k and A represent the thermal conduction coefficient and area, respectively; the subscript *sw*, *ms* and *g* denote the solid wall, mesh screen and gas, respectively; β is the heat conductivity degradation factor, which is reasonably to be set to 0.1 for the stainless

steel mesh screen[1]. There is no mesh screen in the TBT_1; therefore, the intrinsic internal heat loss in TBT_1 can be given as

$$\dot{Q}_{loss,in}\big|_{TBT_1} = \left[k_{sw}A_{sw}\frac{dT}{dx} + k_g A_g \frac{dT}{dx}\right]\Big|_{TBT_1} \tag{S4}$$

Hence, the intrinsic internal heat loss can be calculated by

$$\dot{Q}_{loss,in} = \dot{Q}_{loss,in}\big|_{REG} + \dot{Q}_{loss,in}\big|_{TBT_1} \tag{S5}$$

The obtained $\dot{Q}_{leak,total}$ and $\dot{Q}_{leak,m}$ are given in Figure S3.

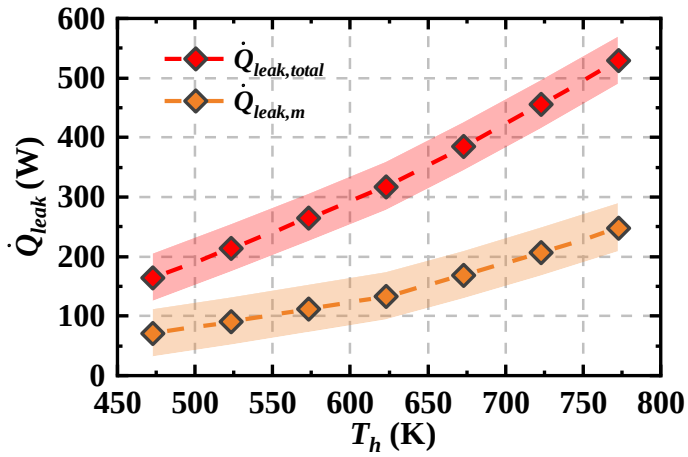


Figure S3. Total heat leak $\dot{Q}_{leak,total}$ and heat leak $\dot{Q}_{leak,m}$ of insulation materials vary with heating temperature T_h. The mean pressure, ambient temperature, and cooling temperature are 5 MPa, 308 K and 280 K, respectively. The colored shaded area represents the error bar.

After obtaining the heat leak of insulation materials by combining Equations (S2-S5), the heating power absorbed by the HHX can be estimated by

$$\dot{Q}_h = \dot{Q}_{h,total} - \dot{Q}_{leak,m} \tag{S6}$$

where $\dot{Q}_{h,total}$ is the heating power measured by the electric power meter.

The cooling power $\dot{Q}_c$ is directly measured by the electric power meter. The energy balance of the solid in CHX indicates that the cooling power (which cools the solid) is equal to the given electric power (which heats the solid) when the solid temperature reaches a stable value. The real cooling power could be even larger due to the heat leakage of the insulation material wrapped around the CHX, which is not considered

with the given small temperature span ($T_0 - T_c = 28$ K) for conservative estimates. Accordingly, the COP can be calculated using Equation (1). Subsequently, the relative Carnot efficiency η_C can be obtained by

$$\eta_C = \frac{\mathrm{COP}}{\mathrm{COP}_C} = \mathrm{COP}\frac{\frac{T_0}{T_c} - 1}{1 - \frac{T_0}{T_h}} \tag{S7}$$

Error Analysis of Measurement

The measurement uncertainty of operating frequency and electric power (which estimates the heating and cooling power) can be directly obtained by the uncertainty of the corresponding measurement sensor displayed in Table S2.

For COP, it is the ratio of cooling power $\dot{Q}_c$ and heating power $\dot{Q}_h$. Due to $\dot{Q}_c$ and $\dot{Q}_h$ are independent and random, the uncertainty of COP can be estimated by[2]

$$\frac{\delta \mathrm{COP}}{\mathrm{COP}} = \sqrt{(\frac{\delta \dot{Q}_c}{\dot{Q}_c})^2 + (\frac{\delta \dot{Q}_h}{\dot{Q}_h})^2} \tag{S8}$$

Simulation Method

This study employs the SAGE program[3] to conduct the numerical investigation (shown in Figure 1C), as well as the design and optimization of the present experimental setup (listed in Table S1). SAGE is widely used for the simulation and design of thermoacoustic systems[4-6]. The optimization is to achieve a highest COP. The simulation model employs three equations of fluid mechanics, i.e., the continuity, momentum, and energy equations, which are outlined below[7]

$$\frac{\partial \rho A_f}{\partial t} + \frac{\partial \rho u A_f}{\partial x} = 0 \tag{S9}$$

$$\frac{\partial \rho u A_f}{\partial t} + \frac{\partial \rho u^2 A_f}{\partial x} + \frac{\partial p}{\partial x} A_f - F A_f = 0 \tag{S10}$$

$$\frac{\partial \rho e A_f}{\partial t} + p\frac{\partial A_f}{\partial t} + \frac{\partial}{\partial x}(\rho u e A_f + u p A_f + \dot{q}) - \dot{Q}_{cv} = 0 \tag{S11}$$

where ρ, u and e denote the density, velocity, and internal energy per unit mass of the gas, respectively; F denotes the viscous terms in the Stokes stress tensor and A_f denotes the flow area of the channel; p, $\dot{q}$ and $\dot{Q}_{cv}$ are the oscillating pressure, axial heat flow and heat flow per unit length due to the convective heat transfer, respectively; t denotes time and x denotes axial dimension. The terms F, q and $\dot{Q}_{cv}$ are given as follows

$$F = -(f_D/d_h + K_{los}/l)\rho u|u|/2 \tag{S12}$$

$$q = -N_k k A_f \frac{\partial T}{\partial x} \tag{S13}$$

$$\dot{Q}_{cv} = Nu(k/d_h)S_x(T_s - T) \tag{S14}$$

where f_D, d_h and K_{los} denote the Darcy friction factor, hydraulic diameter and total local loss coefficient, respectively; N_k is the ratio of the effective gas conductivity to the molecular conductivity; Nu, k, S_x and T_s-T are the Nusselt number, gas thermal conductivity, wetted perimeter and temperature difference between the solid and gas. f_D, Nu and N_k are empirically calculated in Sage software, shown in Table S3.

Table S3. Empirical expressions of f_D, Nu and N_k for the heat exchangers and regenerators in Sage[7]. *Re*, *Pr*, and *Pe* denote the Reynolds number (particularly, *Rt* denotes the turbulent Reynolds number), Prandtl number, and Peclet number, respectively. ∅ denotes the porosity and *ε* denotes the relative roughness. *a* is the ratio of the fin width to the fin length.

Component	Type	f_D	Nu	N_k
Heat exchanger	Shell tube	Laminar: $f_D = \frac{64}{Re}$ Turbulent: $f_D = 0.11(\frac{\varepsilon}{d_h} + \frac{68}{Re})^{0.25}$	Laminar steady: $Nu = 6$ Turbulent: $Nu = 0.036Re^{0.8}(\frac{l}{d_h})^{-0.055}Pr^{0.33}$	Laminar: $N_k = 1$ Turbulent: $N_k = 0.022Re^{0.75}Pr$

	Plate fin	Laminar: $f_D = \frac{96}{Re}$ Turbulent: $f_D = 0.11(\frac{\varepsilon}{d_h} + \frac{68}{Re})^{0.25}$	Laminar steady: $Nu = 10c$ where $c = 0.438 + 0.562(1-a)^3$ Turbulent: $Nu = 0.035Re^{0.75}Pr^{0.33}$	Laminar: $N_k = 1$ Turbulent: $N_k = 0.022Re^{0.75}Pr$
Regenerator	Woven screen matrix	$f_D = \frac{129}{Re} + 2.91Re^{-0.103}$	$Nu = (1 + 0.99Pe^{0.66})\phi^{1.79}$	$N_k = 1 + 0.5Pe^{0.66}\phi^{-2.91}$
Cylinder		$f_D = 0$	$Nu = 6N_k$	$N_k = 1 + 0.014Rt$
	Tube	Laminar: $f_D = \frac{64}{Re}$ Turbulent: $f_D = 0.11(\frac{\varepsilon}{d_h} + \frac{68}{Re})^{0.25}$	Laminar steady: $Nu = 6$ Turbulent: $Nu = 0.036Re^{0.8}(\frac{l}{d_h})^{-0.055}Pr^{0.33}$	Laminar: $N_k = 1$ Turbulent: $N_k = 0.022Re^{0.75}Pr$

The water in the liquid resonator is considered as incompressible liquid, thus can be simulated using the Reciprocator module in Sage due to the incompressible liquid approximation. The equivalent stiffness coefficient K and damping coefficient R_m for the liquid can be calculated using the following equations[8]:

$$K = \frac{\rho_{liq} g \pi {D_{liq}}^2}{2} \tag{S15}$$

$$R_m = \pi^{1.5} D_{liq} l_{liq} \sqrt{\rho_{liq} \mu_{liq} f} + 0.84\pi^2 \rho_{liq} f X_{liq} (D_{liq}/2)^2 E \tag{S16}$$

where the subscript *liq* denotes the water; g, D, l μ denote the gravitational acceleration, diameter, length and dynamic viscosity, respectively; X is the displacement amplitude and E is a coefficient estimating the total minor loss.

The total energy flow $\dot{H}_2$ can be expressed in frequency domain based on the classical thermoacoustic theory[9], and the heat exchange $\dot{Q}$ of the heat exchangers (AHX, HHX, and CHX) can be calculated by the difference in $\dot{H}_2$ between the outlet and inlet. The expressions are as follows

$$\dot{H}_2 = \frac{1}{2}p_1\widetilde{U}_1\left[1-\frac{\left(f_\kappa-\widetilde{f}_v\right)}{(1-\widetilde{f}_v)(1+Pr)}\right]+\frac{\rho_m c_p|U_1|^2}{2A_f\omega(1-Pr)^2|1-f_v|^2}\mathrm{Im}[f_\kappa + \widetilde{f}_v Pr]\frac{dT_m}{dx}-(A_g k_g + k_{sw}A_{sw} + \beta k_{ms}A_{ms}\frac{dT}{dx})\frac{dT_m}{dx} \tag{S17}$$

$$\dot{Q} = \dot{H}_{2,out} - \dot{H}_{2,in} \tag{S18}$$

where the subscripts *in* and *out* denote the inlet and outlet of the heat exchanger, respectively; ω denote the angular frequency; f_κ and f_v are the spatially averaged viscous and thermal functions reflecting the flow and heat transfer, respectively, determined by the geometric structure of the flow channel and the property of the working gas. For the plate-fin heat exchanger (i.e., HHX and CHX), the expressions are as follows:

$$f_{v,\kappa} = \frac{\tanh\left[(1+i)y_0/\delta_{v,\kappa}\right]}{(1+i)y_0/\delta_{v,\kappa}} \tag{S19}$$

where y_0 is half of the fin spacing, δ_v and δ_κ are the viscous and thermal penetration depths of the gas, respectively. While for the shell-tube heat exchanger (i.e., AHX), the corresponding expressions are:

$$f_{v,\kappa} = \frac{2J_1[(i-1)r/\delta_{v,\kappa}]}{J_0[(i-1)r/\delta_{v,\kappa}](i-1)r/\delta_{v,\kappa}} \tag{S20}$$

where J_0 and J_1 are the zero-order and first-order Bessel functions, respectively, r is the radius of the heat-exchange tube. The δ_v and δ_κ are expressed as follows

$$\delta_v = \sqrt{2\nu/\omega} \tag{S21}$$

$$\delta_\kappa = \sqrt{2\kappa/\omega} \tag{S22}$$

where ν and κ are the kinematic viscosity and thermal diffusivity of the gas, respectively.

The expressions of r_v and r_k (the viscous and thermal resistance per unit length, respectively), are given by

$$r_v = \frac{\omega \rho_m}{A_f} \frac{\mathrm{Im}[-f_v]}{|1 - f_v|^2} \tag{S23}$$

$$\frac{1}{r_\kappa} = \frac{\gamma - 1}{\gamma} \frac{\omega A_f}{p_m} \mathrm{Im}[-f_\kappa] \tag{S24}$$

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